IMPROPER INTEGRALS

Converging and diverging, side by side

2
COMPARED

The same kind of integral, two outcomes: it all depends on how fast the integrand decays.

	CONVERGES	DIVERGES
THE DEFINING LIMIT	exists and is finite	is infinite or does not exist
EXAMPLE	$\lim_{b\to\infty}\left(1-\frac{1}{b}\right)=1$	$\lim_{b\to\infty}\ln b=\infty$
INTEGRAND AT ONE HUNDRED	$\frac{1}{x^2}=0.0001$	$\frac{1}{x}=0.01$, a hundred times larger
DECAY	fast enough	not fast enough, although $\frac{1}{x}\to 0$

The ratio between the two integrands is x itself, so the gap keeps widening as x grows.

A vanishing integrand is not enough: $\frac{1}{\sqrt{x}}$ also tends to 0, and its integral diverges too.

The p-test turns this comparison into a rule for every power of x.